

Agent Experience

Original query q_i

four yellow candles
in front of three
sparkling birds to the
right of a chair

$\rightarrow \pi_\theta \rightarrow$

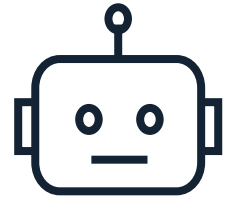


Original model

Text-to-Image Agent

Plan · Retrieve skills · Refine

$\mathcal{H}_r^\star$ using π_θ



Harness prompt p_i

A weathered wooden chair stands
at the back-left. Exactly three
sparkling songbirds stand to its
right. Exactly four yellow beeswax
candles occupy the foreground,
directly in front of the birds.

$\xrightarrow{\pi_\theta}$

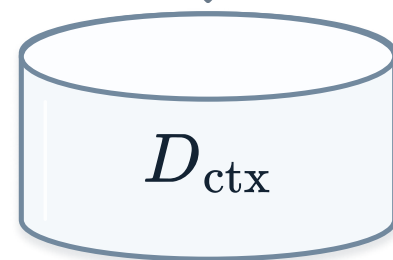


Selected image

I_i

Context records

$\{(q_i, p_i, I_i)\}$



D_{ctx}

Diffusion On-policy Context Distillation

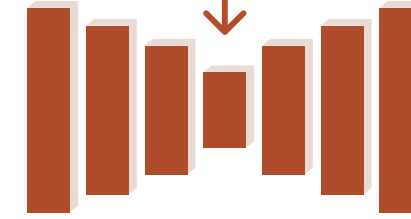
(q_i, p_i) from D_{ctx}

Teacher

EMA weights

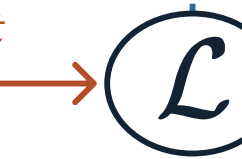
Original query + harness prompt

$c_i^t = F_T([q_i; p_i])$



$\nu_{i,k}^t = v_\theta(x_{i,t_k}^s, t_k, c_i^t)$

stop-gradient



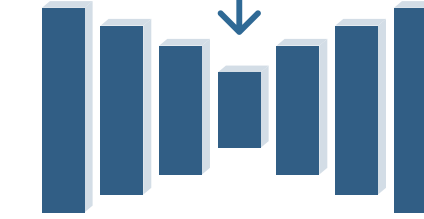
Velocity matching

Student

Trainable weights

Original query only

$c_i^s = F_T(q_i)$



$\nu_{i,k}^s = v_\theta(x_{i,t_k}^s, t_k, c_i^s)$

retain student

EMA update

gradient

Student on-policy rollout



$x_{i,t_K}^s \sim \mathcal{N}(0, \mathbf{I})$

...



x_{i,t_k}^s

...



x_{i,t_0}^s

(x_{i,t_k}^s, t_k)
conditioned on q only

$$\mathcal{L}_{\text{D-OPCD}} = \mathbb{E}_{D_{\text{ctx}}} \left[\frac{1}{K} \sum_{k=1}^K \|\nu_{i,k}^s - \text{sg}(\nu_{i,k}^t)\|_2^2 \right]$$

Query-only Generation

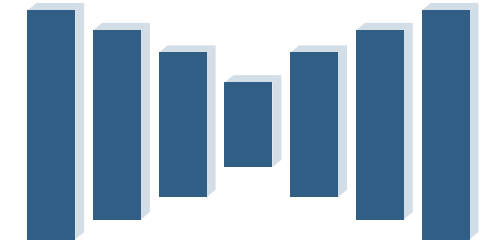
Same original query

q_i

four yellow candles in front of
three sparkling birds to the right
of a chair

Updated generator

$\pi_{\theta'}$



Direct generation from q